

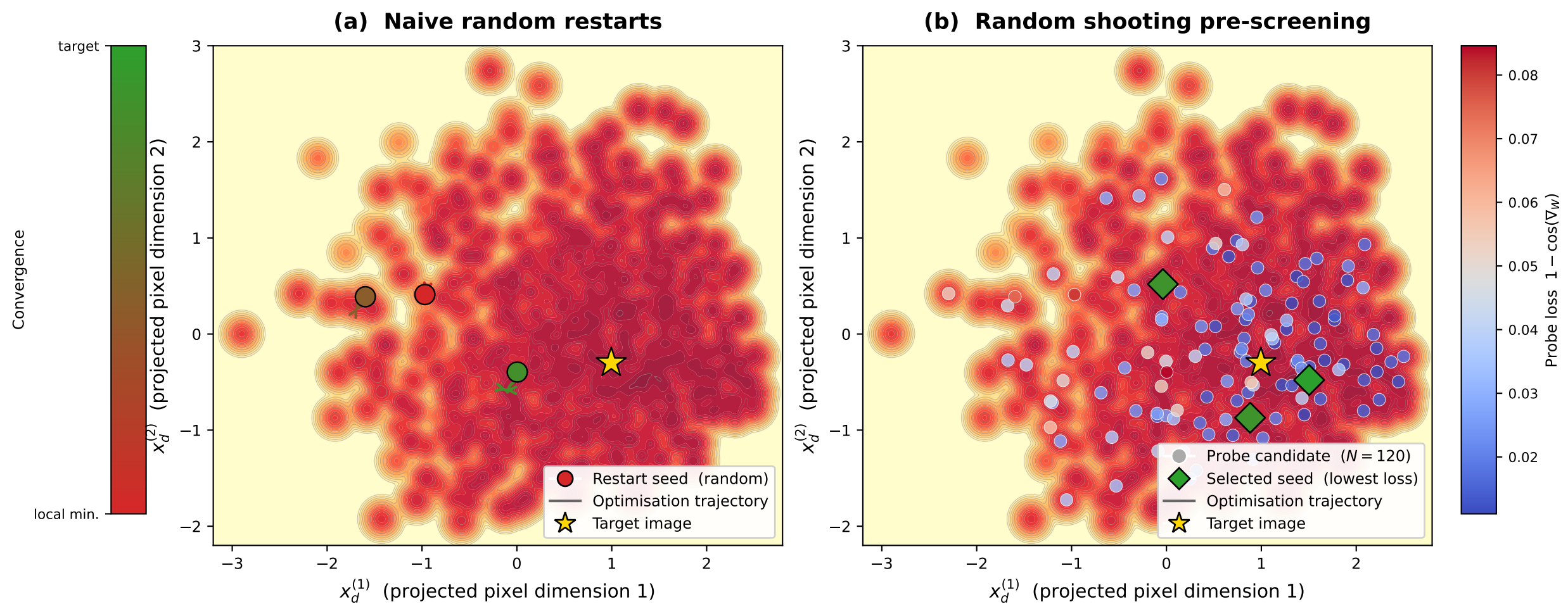


Figure: 2D PCA projection of grayscale LFW faces. Colour shows the true gradient-matching loss $1 - \cos(\nabla_w \mathcal{L}(x_d), \nabla_w \mathcal{L}(x))$ (random LeNet), evaluated by inverting the PCA projection at each point. (a) Naive restarts land in high-loss basins. (b) Random shooting evaluates N probes, selects lowest-loss seeds (diamonds), and steers restarts toward the target image — though not all converge (see convergence scale).